

Linear Algebra & Calculus

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 36

Question 1 of 36 | GATE 2025 Set 1 (CS1) | Q.23 (Computer Science & IT)

Q.23

Consider the given system of linear equations for variables x and y , where k is a real-valued constant. Which of the following option(s) is/are CORRECT?

$$x + ky = 1$$

$$kx + y = -1$$

- (A) There is exactly one value of k for which the above system of equations has no solution.
- (B) There exist an infinite number of values of k for which the system of equations has no solution.
- (C) There exists exactly one value of k for which the system of equations has exactly one solution.
- (D) There exists exactly one value of k for which the system of equations has an infinite number of solutions.

Question 2 of 36 | GATE 2025 Set 1 (CS1) | Q.31 (Computer Science & IT)

Q.31

Consider the given function $f(x)$.

$$f(x) = \begin{cases} ax + b & \text{for } x < 1 \\ x^3 + x^2 + 1 & \text{for } x \geq 1 \end{cases}$$

$$x^3 + x^2 + 1 \text{ for } x \geq 1$$

If the function is differentiable everywhere, the value of b must be _____.

(rounded off to one decimal place)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 3 of 36 | GATE 2025 Set 1 (CS1) | Q.39 (Computer Science & IT)

Q.39

Consider two relations describing teams and players in a sports league:

- teams(tid, tname): tid, tname are team-id and team-name, respectively
- players(pid, pname, tid): pid, pname, and tid denote player-id, player-name and the team-id of the player, respectively

Which ONE of the following tuple relational calculus queries returns the name of the players who play for the team having tname as 'MI'?

- (A) $\{ p.pname \mid p \sqsubseteq \text{players} \sqcap t (t \sqsubseteq \text{teams} \sqcap p.tid = t.tid \sqcap t.tname = 'MI') \}$
- (B) $\{ p.pname \mid p \sqsubseteq \text{teams} \sqcap t (t \sqsubseteq \text{players} \sqcap p.tid = t.tid \sqcap t.tname = 'MI') \}$
- (C) $\{ p.pname \mid p \sqsubseteq \text{players} \sqcap t (t \sqsubseteq \text{teams} \sqcap t.tname = 'MI') \}$
- (D) $\{ p.pname \mid p \sqsubseteq \text{teams} \sqcap t (t \sqsubseteq \text{players} \sqcap t.tname = 'MI') \}$

Question 4 of 36 | GATE 2025 Set 1 (CS1) | Q.41 (Computer Science & IT)

Q.41

Let A be a 2×2 matrix as given.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

1

1

-1]

What are the eigenvalues of the matrix A?

- (A) 1, -1
- (B) $2\sqrt{2}$, $-2\sqrt{2}$
- (C) $4\sqrt{2}$, $-4\sqrt{2}$
- (D) $64\sqrt{2}$, $-64\sqrt{2}$

Question 5 of 36 | GATE 2025 Set 2 (CS2) | Q.12 (Computer Science & IT)

Q.12

The value of x such that $x > 1$, satisfying the equation $\int_1^x \ln t \, dt =$

1

4

x

1

is

- (A) $\sqrt{e}$
- (B) e
- (C) e^2
- (D) e^{-1}

Question 6 of 36 | GATE 2025 Set 2 (CS2) | Q.14 (Computer Science & IT)

Q.14

Let L, M, and N be non-singular matrices of order 3 satisfying the equations

$$L^2 = L^{-1}, M = L^8 \text{ and } N = L^2.$$

Which ONE of the following is the value of the determinant of (M-N)?

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Question 7 of 36 | GATE 2025 Set 2 (CS2) | Q.44 (Computer Science & IT)

Q.44

Consider a system of linear equations $PX = Q$ where $P \in \mathbb{R}^{3 \times 3}$ and $Q \in \mathbb{R}^{3 \times 1}$.

Suppose P has an LU decomposition, $P = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$$

1

0

0

 l_{21}

1

0

 l_{31} l_{32}

1

] and $U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$. u_{11} u_{12} u_{13}

0

 u_{22} u_{23}

0

0

 u_{33}

].

Which of the following statement(s) is/are TRUE?

- (A) The system $PX = Q$ can be solved by first solving $LY = Q$ and then $UX = Y$.
- (B) If P is invertible, then both L and U are invertible.
- (C) If P is singular, then at least one of the diagonal elements of U is zero.
- (D) If P is symmetric, then both L and U are symmetric.

Question 8 of 36 | GATE 2024 Set 1 (CS1 S5) | Q.11 (Computer Science & IT)

Q.11

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = \max\{x, x^3\}$, $x \in \mathbb{R}$, where $\mathbb{R}$ is the

set

of

all

real

numbers.

The

set
of
all
points
where
 $f(x)$
is
NOT differentiable is

- (A) $\{-1, 1, 2\}$
- (B) $\{-2, -1, 1\}$
- (C) $\{0, 1\}$
- (D) $\{-1, 0, 1\}$

Question 9 of 36 | GATE 2024 Set 1 (CS1 S5) | Q.12 (Computer Science & IT)

Q.12

The product of all eigenvalues of the matrix [

1
2
3
4
5
6
7
8
9
] is

- (A) -1
 - (B) 0
 - (C) 1
 - (D) 2
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Question 10 of 36 | GATE 2024 Set 1 (CS1 S5) | Q.49 (Computer Science & IT)

Q.49

Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = 0$?

- (A) There exist at least $m-n$ linearly independent solutions to this system
- (B) There exist $m-n$ linearly independent vectors such that every solution is a linear combination of these vectors

- (C) There exists a non-zero solution in which at least $m-n$ variables are 0
- (D) There exists a solution in which at least n variables are non-zero

Question 11 of 36 | GATE 2024 Set 2 (CS2 S6) | Q.16 (Computer Science & IT)

Q.16

Let $f(x)$ be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$f(x) = 1 - f(2 - x)$$

Which one of the following options is the CORRECT value of $\int f(x) dx$

2
0
?

- (A) 0
- (B) 1
- (C) 2
- (D) -1

Question 12 of 36 | GATE 2024 Set 2 (CS2 S6) | Q.17 (Computer Science & IT)

Q.17

Let A be the adjacency matrix of a simple undirected graph G . Suppose A is its own inverse. Which one of the following statements is always TRUE?

- (A) G is a cycle
- (B) G is a perfect matching
- (C) G is a complete graph
- (D) There is no such graph G

Question 13 of 36 | GATE 2024 Set 2 (CS2 S6) | Q.47 (Computer Science & IT)

Q.47

Let A be an $n \times n$ matrix over the set of all real numbers $\mathbb{R}$. Let B be a matrix obtained from A by swapping two rows. Which of the following statements is/are TRUE?

- (A) The determinant of B is the negative of the determinant of A
- (B) If A is invertible, then B is also invertible
- (C) If A is symmetric, then B is also symmetric
- (D) If the trace of A is zero, then the trace of B is also zero

Question 14 of 36 | GATE 2023 Single Session | Q.17 (Computer Science & IT)

Q.17

Let

 $A =$

1

2

3

4

4

1

2

3

3

4

1

2

2

3

4

1

and

 $B =$

3

4

1

2

4

1

2

3

1

2

3

4

2

3

4

1

.

Let $\det(A)$ and $\det(B)$ denote the determinants of the matrices A and B , respectively.

Which one of the options given below is TRUE?

- (A) $\det(A) = \det(B)$
- (B) $\det(B) = -\det(A)$
- (C) $\det(A) = 0$
- (D) $\det(AB) = \det(A) + \det(B)$ CS

Question 15 of 36 | GATE 2023 Single Session | Q.27 (Computer Science & IT)

Q.27

Let

$$f(x) = x^3 + 15x^2 - 33x - 36$$

be a real-valued function.

Which of the following statements is/are TRUE?

- (A) $f(x)$ does not have a local maximum.
- (B) $f(x)$ has a local maximum.
- (C) $f(x)$ does not have a local minimum.
- (D) $f(x)$ has a local minimum. CS

Question 16 of 36 | GATE 2023 Single Session | Q.29 (Computer Science & IT)

Q.29

Let A be the adjacency matrix of the graph with vertices $\{1, 2, 3, 4, 5\}$.

1
2
4
3
5

Let $\lambda_1, \lambda_2, \lambda_3, \lambda_4$, and λ_5 be the five eigenvalues of A. Note that these eigenvalues need not be distinct.

The value of $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 =$

.

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 17 of 36 | GATE 2023 Single Session | Q.30 (Computer Science & IT)

Q.30

The value of the definite integral

3
-3
2
-2

1
 -1
 $4x^2y - z^3$
 $dz \, dy \, dx$
 is
 . (Rounded off to the nearest integer)
 CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 18 of 36 | GATE 2022 Single Session | Q.20 (Computer Science & IT)

Q.20

Consider the following two statements with respect to the matrices

$m \times n$
 A ,
 $n \times m$
 B ,
 $n \times n$
 C and

D .

Statement 1: $\text{tr}(AB) = \text{tr}(BA)$

Statement 2: $\text{tr}(CD) = \text{tr}(DC)$

where $\text{tr}()$ represents the trace of a matrix. Which one of the following holds?

- (A) Statement 1 is correct and Statement 2 is wrong.
- (B) Statement 1 is wrong and Statement 2 is correct.
- (C) Both Statement 1 and Statement 2 are correct.
- (D) Both Statement 1 and Statement 2 are wrong.

Question 19 of 36 | GATE 2022 Single Session | Q.34 (Computer Science & IT)

Q.34

The value of the following limit is _____.

2
 0^+
 x
 x
 x
 $\lim$
 e
 $\frac{1}{x^2}$

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 20 of 36 | GATE 2022 Single Session | Q.37 (Computer Science & IT)

Q.37

Consider a simple undirected unweighted graph with at least three vertices. If A is the adjacency matrix of the graph, then the number of 3-cycles in the graph is given by the trace of

- (A) $3A$
- (B) $3A$ divided by 2
- (C) $3A$ divided by 3
- (D) $3A$ divided by 6

Question 21 of 36 | GATE 2022 Single Session | Q.52 (Computer Science & IT)

Q.52

Which of the properties hold for the adjacency matrix A of a simple undirected unweighted graph having n vertices?

- (A) The diagonal entries of $2A$ are the degrees of the vertices of the graph.
- (B) If the graph is connected, then none of the entries of $\frac{1}{n}A$ can be zero.
- (C) If the sum of all the elements of A is at most $2(n-1)$, then the graph must be acyclic.
- (D) If there is at least a 1 in each of A 's rows and columns, then the graph must be connected.

Question 22 of 36 | GATE 2022 Single Session | Q.53 (Computer Science & IT)

Q.53

Which of the following is/are the eigenvector(s) for the matrix given below?

- 9
- 6
- 2
- 4
- 8
- 6
- 3
- 1
- 20
- 15
- 8
- 5
- 32

21
7
12
☐
☐
☐
☐
☐
☐
☐
☐
☐
☐
☐

- (A) 1 1 0 1 ☐ ☐ ☐
(B) 1 0 1 0 ☐ ☐ ☐
(C) 1 0 2 2 ☐ ☐ ☐
(D) 0 1 3 0 ☐ ☐ ☐

Question 23 of 36 | GATE 2022 Single Session | Q.58 (Computer Science & IT)

Q.58

Let

(,
)

$G(V, E)$ be a directed graph, where

$V = \{1, 2, 3, 4, 5\}$

V is

the set of vertices and

E is the set of directed edges, as defined by the following adjacency matrix

.

A

$A[i][j] = 1, 1 \leq j \leq 5$

0, otherwise

$A[i][j] = 1$ indicates a directed edge from node i to node j . A directed spanning tree of

,

G rooted at

,

r

V

T

is defined as a subgraph T of G such that the undirected

version of T is a tree, and T contains a directed path from r to every other vertex in

V The number of such directed spanning trees rooted at vertex 5 is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 24 of 36 | GATE 2019 Single Session | Q.9 (Computer Science & IT)

Q.9

Let X be a square matrix. Consider the following two statements on X .

I.

X is invertible.

II.

Determinant of X is non-zero.

Which one of the following is TRUE?

- (A) I implies II; II does not imply I.
- (B) II implies I; I does not imply II.
- (C) I does not imply II; II does not imply I.
- (D) I and II are equivalent statements.

Question 25 of 36 | GATE 2019 Single Session | Q.13 (Computer Science & IT)

Q.13 Compute $\lim_{x \rightarrow 3} \frac{x^4 - 81}{2x^2 - 5x - 3}$

$x \rightarrow 3$

$x^4 - 81$

$2x^2 - 5x - 3$

- (A) 1
- (B) $53/12$
- (C) $108/7$
- (D) Limit does not exist

Question 26 of 36 | GATE 2019 Single Session | Q.44 (Computer Science & IT)

Q.44 Consider the following matrix:

$R = \begin{bmatrix} 1 & 2 & 4 & 8 \end{bmatrix}$

1

2

4

8

1
3
9
27
1
4
16
64
1
5
25
125
]

The absolute value of the product of Eigen values of R is .

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 27 of 36 | GATE 2018 Single Session | Q.16 (Computer Science & IT)

Q.16 The value of $\int_0^{\pi/4} x \cos(x^2) dx$
 $\pi/4$
0
correct to three decimal places
(assuming that $\pi = 3.14$) is ____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 28 of 36 | GATE 2018 Single Session | Q.17 (Computer Science & IT)

Q.17
Consider a matrix
T
A
uv
 $\square$
where
1
1
,
2
1
u
v
 $\square$

□
□
□
□
□

. Note that

T_v denotes the

transpose of v . The largest eigenvalue of A is ____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 29 of 36 | GATE 2018 Single Session | Q.26 (Computer Science & IT)

Q.26

Consider a matrix P whose only eigenvectors are the multiples of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

□□

.

Consider the following statements.

(I) P does not have an inverse

(II) P has a repeated eigenvalue

(III) P cannot be diagonalized

Which one of the following options is correct?

- (A) Only I and III are necessarily true
- (B) Only II is necessarily true
- (C) Only I and II are necessarily true
- (D) Only II and III are necessarily true CS 8/20

Question 30 of 36 | GATE 2018 Single Session | Q.31 (Computer Science & IT)

Q.31 Assume that multiplying a matrix G_1 of dimension $p \times q$ with another matrix G_2 of dimension $q \times r$ requires pqr scalar multiplications. Computing the product of n matrices $G_1 G_2 G_3 \dots G_n$ can be done by parenthesizing in different ways. Define $G_i G_{i+1}$ as an explicitly computed pair for a given paranthesization if they are directly multiplied. For example, in the

matrix

multiplication

chain

$G_1 G_2 G_3 G_4 G_5 G_6$

using

parenthesization

$(G_1(G_2 G_3))(G_4(G_5 G_6))$, $G_2 G_3$ and $G_5 G_6$ are the only explicitly computed pairs.

Consider a matrix multiplication chain $F_1 F_2 F_3 F_4 F_5$, where matrices F_1, F_2, F_3, F_4 and F_5 are

of dimensions 2×25 , 25×3 , 3×16 , 16×1 and 1×1000 , respectively. In the parenthesization of $F1F2F3F4F5$ that minimizes the total number of scalar multiplications, the explicitly computed pairs is/are

- (A) $F1F2$ and $F3F4$ only
- (B) $F2F3$ only
- (C) $F3F4$ only
- (D) $F1F2$ and $F4F5$ only

Question 31 of 36 | GATE 2016 Set 1 (CS1) | Q.5 (Computer Science & IT)

Q.5

Two eigenvalues of a 3×3 real matrix P are $(2 + \sqrt{-1})$ and 3 . The determinant of P is

.

CS(Set A)

1/17

GATE 2016

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 32 of 36 | GATE 2016 Set 1 (CS1) | Q.38 (Computer Science & IT)

Q.38

Consider the weighted undirected graph with 4 vertices, where the weight of edge $\{i, j\}$ is given by the entry W_{ij} in the matrix W .

$W =$

$\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$

0

2

8

5

2

0

5

8

8

5

0

x

5

8

x

0

$\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$

□□□

The largest possible integer value of x , for which at least one shortest path between some pair of vertices will contain the edge with weight x is

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 33 of 36 | GATE 2016 Set 2 (CS2) | Q.2 (Computer Science & IT)

Q.2

Let $f(x)$ be a polynomial and $g(x) = f'(x)$ be its derivative. If the degree of $(f(x) + f(-x))$ is 10, then the degree of $(g(x) - g(-x))$ is

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 34 of 36 | GATE 2016 Set 2 (CS2) | Q.4 (Computer Science & IT)

Q.4

Consider the systems, each consisting of m linear equations in n variables.

- I. If $m < n$, then all such systems have a solution
- II. If $m > n$, then none of these systems has a solution
- III. If $m = n$, then there exists a system which has a solution

Which one of the following is CORRECT?

- (A) I, II and III are true
- (B) Only II and III are true
- (C) Only III is true
- (D) None of them is true CS(Set B) 1/18 GATE 2016

Question 35 of 36 | GATE 2016 Set 2 (CS2) | Q.6 (Computer Science & IT)

Q.6

Suppose that the eigenvalues of matrix A are 1, 2, 4.

The determinant of $(A - I)^T$ is

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 36 of 36 | GATE 2016 Set 2 (CS2) | Q.38 (Computer Science & IT)

Q.38

Let A_1, A_2, A_3 , and A_4 be four matrices of dimensions $10 \times 5, 5 \times 20, 20 \times 10$, and 10×5 , respectively. The minimum number of scalar multiplications required to find the product $A_1 A_2 A_3 A_4$ using the basic matrix multiplication method is

CS(Set B)

10/18

GATE 2016

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]
